Shashank Kamble

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Summary

Master's student in IT & Analytics (AI Concentration) at Rutgers University pursuing **AI/ML Engineer** and **Software Developer** roles, with experience building and evaluating RAG pipelines, fine-tuning LLMs, and shipping full-stack data applications. Skilled in Python, HuggingFace Transformers, PostgreSQL, and Docker, with a track record of taking projects from data pipeline to deployed, user-facing product.

Education

Rutgers University Aug 2025 – Present
M.S. IT & Analytics (AI Concentration) | GPA: 3.66
Relevant Coursework: LLMs, Machine Learning, Data Management, Business Analytics Programming

St. Francis Institute of Technology June 2021 – May 2025
B.E. Computer Science (Cyber Security Honors) | CGPA: 8.57/10

Skills

Programming: Python, SQL, R, C
ML & NLP: LLM Fine-Tuning, RAG Pipelines, Retrieval (BM25, Dense, SPLADE), Transformer Fine-Tuning, Multilingual NLP, Model Evaluation
Frameworks & Libraries: HuggingFace Transformers, scikit-learn, Pandas, NumPy, OpenCV, MediaPipe
Software Engineering: Full-Stack Development, PostgreSQL, Streaming Data Processing, RESTful APIs
Deployment & Tools: Docker, Vercel, Git/GitHub, Cursor, Claude Code, GitHub Copilot

Experience

Supply Chain & Workforce Analytics – Moxie Salon & Beauty Bar Sept 2026 – Present
Academic Client Engagement – Rutgers Business School Analytics Practicum

- Built a data pipeline integrating Zenoti POS, payroll, and QuickBooks data across 3 locations (∼45 employees) into unified KPI benchmarks for downstream modeling.
- Engineered feature pipelines (RPLH, labor cost %, utilization, walk-in mix) to surface anomalies, detecting a ∼12% labor cost overrun at one location.
- Developing a demand-to-labor forecasting model (time-series/regression-based) projected to reduce overstaffing hours by ∼15%, translating seasonality signals into staffing predictions.

All India Radio (AIR) June 2023 – July 2023
Research & Programming Intern

- Built Python scripts to process and analyze broadcast/audience datasets across 10+ programs, extracting 3 engagement patterns via statistical analysis.
- Automated a manual reporting workflow, reducing weekly processing time by ∼20%.

Projects

RAG-based Housing Recommendation System (NJ) | Python, FAISS, SPLADE, LLMs

- Benchmarked 5 retrieval pipelines (BM25, Dense, Hybrid, Query Rewriting, SPLADE-v3 + Reranker) across 7 NJ public-service domains using a manually curated 70-question evaluation benchmark
- Improved Recall@5 from 0.52 to 0.69 using SPLADE-v3 with Qwen3-Reranker-4B, the strongest performer across all evaluated methods
- Built category-aware FAISS/SPLADE indexes over 46K+ structured chunks, reducing hallucination rate to 2.86% while achieving 76.43% answer accuracy via grounded generation with Qwen2.5-32B-Instruct

LLM Response Evaluation System (Therapeutic Domain) | Python, HuggingFace, SBERT

- Fine-tuned and evaluated 5 open-source LLMs on a 600-sample psychotherapy dataset spanning 13 therapeutic techniques
- Built a multi-axis evaluation framework combining SBERT similarity, sentiment analysis, and human validation to benchmark conversational quality
- Identified Falcon and Gemini as strongest performers for therapeutic coherence through embedding-based evaluation and PCA analysis

Wrist-Wise – Health Analytics Platform | Python, PostgreSQL, Full-Stack

- Built a full-stack health analytics platform processing large Apple Health XML exports (up to 600MB) via streaming parsing and batch processing, with PostgreSQL for persistent storage
- Designed an interactive dashboard with drag-and-drop file uploads and real-time refresh, plus a derived 0–100 recovery score from activity, calorie, and heart-rate data
- Implemented personalized trend analysis surfacing 30-day patterns and generating workout recommendations from historical health data

Certifications

Google Data Analytics Professional Certificate | AWS Cloud Practitioner | Machine Learning – University of London | Python for Data Science – University of Michigan